Seven of Eight AI Deployment Patterns Are Structurally Suppressed

The 2026 landscape propagates on capital, distribution, and lock-in — not on structural merit.

THE GROVE FOUNDATION · AI PATTERN BENCHMARK · MARCH 2026 ·

What this benchmark measures

Gartner measures market momentum. Forrester measures vendor capability. IDC measures competitive positioning. None of them asks the question that determines whether a technology wins the 36-month race: does the pattern propagate on its own structural merits, or does it require continuous external subsidy to sustain adoption?

The Λ framework measures that question directly. It treats adoption as a structural mechanics problem. Capital can accelerate inferior architectures for three to five years. Distribution power can sustain them inside an installed base. Neither subsidy survives the displacement pressure that accumulates once cognitive friction drops and an alternative becomes structurally accessible.

Eight AI deployment patterns. One equation. Observable data only — legal frameworks, deployment economics, regulatory exposure, cognitive load. Ninety-six cited sources. No vendor narratives.

The finding

One pattern is Approaching Critical: Mistral/DeepSeek at $\Lambda = 0.0314$. Two patterns sit Sub-Critical. Five patterns — including all three platform-bundle and centralized-API incumbents — are Structurally Inert.

The $650 billion AI infrastructure buildout is structurally suppressed. It propagates on venture capital, enterprise distribution power, and accumulated contractual lock-in. Remove the subsidy and the propagation stalls or reverses.

That is the finding. The rest of this paper shows the math.

The equation, briefly

$$\Lambda = (S \times R \times V) \cdot \left[1 / (1 + (\beta \cdot Fc)^2) \right]$$

Spreadability (S) — how freely the pattern copies. Licensing permissiveness, network topology, replication cost. Range 0 to 1.

Standardized Rails (R) — how easily the pattern integrates with existing infrastructure. Protocol compatibility, deployment tooling maturity, ecosystem support. Range 0 to 1.

Validation Multiplier (V) — a discount applied to theoretical architectures without deployment data. Range 0.2 (pre-publication) to 1.0 (validated at enterprise scale). New in methodology version 2.0. Prevents theoretical patterns from scoring at theoretical maximums.

Cognitive Friction (Fc) — resistance to adoption measured by objectively observable task-switching costs. Documentation length, UI complexity, paradigm-shift demands, operational process changes. Range 0 to 10.

Exogenous Incentive (β) — external forcing functions that accelerate or dampen adoption. Geometric mean of financial, regulatory, and ideological incentive sub-scores. Lower values indicate stronger forcing functions. Range 0.1 to 10.

The equation establishes that base structural strength (S × R × V) is necessary but not sufficient. Friction and incentive dominate whenever their product becomes large. A pattern with excellent rails and poor incentives does not propagate. A pattern with moderate rails and powerful incentives — Bitcoin is the canonical historical case — can overcome extreme friction.

Four historical cases anchor the framework

The Λ framework is not arbitrary. Calibrated against four well-documented technology adoptions, it reproduces recognizable outcomes.

Case	S	R	V	Fc	β	Λ	Tier
TCP/IP (vs. OSI)	0.95	0.80	1.0	2.5	0.33	**0.452**	Critical Mass
ISO Shipping Container	0.80	0.95	1.0	1.0	1.00	**0.380**	Critical Mass
Bitcoin	0.90	0.40	1.0	7.0	0.20	**0.122**	Critical Mass
US Metric System	0.80	0.40	1.0	10.0	1.00	**0.003**	Structurally Inert

TCP/IP propagated at Λ = 0.452. Open standard. Permissive replication. Powerful interconnection incentive overcame moderate technical friction.

The ISO Shipping Container propagated at Λ = 0.380. Near-zero cognitive friction. Immediate operational compatibility with existing port infrastructure.

Bitcoin propagated at Λ = 0.122 despite Fc = 7.0 — extreme cognitive friction. The counterweight was β = 0.20, an unprecedented speculative financial incentive.

The US Metric System has remained at Λ = 0.003 — Structurally Inert — in the American consumer market for 150 years.

Each case passes the intuition test. The math produces the history.

APRIL STRUCTURAL SHIFT — Note on Gemma 4 and Open-Weight Dynamics

The data and Λ scoring in this March 2026 benchmark were finalized prior to the early April release of Gemma 4 under an Apache license. The release represents a significant advancement on the open-weight front and materially alters the structural mechanics of the landscape. Full mathematical impact will be reflected in the Q2 landscape update, scheduled for late June.

The 2026 landscape, scored

Pattern	Category	S	R	V	Fc	β	S×R×V	Λ	Tier
Mistral / DeepSeek	Open-Weight	0.80	0.60	1.0	6.0	0.630	0.480	**0.0314**	Approaching Critical
Meta Llama	Open-Weight	0.70	0.70	1.0	5.0	1.357	0.490	**0.0104**	Sub-Critical
Apple Intelligence	On-Device	0.20	0.57	1.0	2.0	1.710	0.114	**0.0090**	Sub-Critical
Anthropic Claude	Centralized API	0.47	0.80	1.0	5.0	1.587	0.376	**0.0059**	Sub-Critical
OpenAI GPT	Centralized API	0.47	0.93	1.0	6.0	2.924	0.437	**0.0014**	Structurally Inert
Google Gemini	Platform Bundle	0.30	0.80	1.0	5.0	2.924	0.240	**0.0011**	Structurally Inert
Microsoft Copilot	Platform Bundle	0.30	0.80	1.0	8.0	5.000	0.240	**0.0002**	Structurally Inert
Autonomaton	Sovereign Open	0.93	0.83	0.2	6.0	6.300	0.154	**0.0001**	Structurally Inert

Two structural clusters define the field

The Dependency Profile combines high Standardized Rails with low Spreadability. OpenAI GPT, Google Gemini, Anthropic Claude, and Microsoft Copilot sit in this cluster. These patterns integrate easily. Enterprise infrastructure is already shaped around them. Proprietary constraints suppress the transferability of anything the organization learns while using them. Each quarter of adoption accumulates switching cost rather than structural leverage.

The Sovereignty Profile combines high Spreadability with variable Cognitive Friction. Mistral/DeepSeek, Meta Llama, Apple Intelligence, and the Autonomaton pattern sit in this cluster. These patterns permit transferable competence. The capability builds inside the organization rather than inside the vendor.

Dependency Profile patterns are easier to adopt and harder to leave. Sovereignty Profile patterns are harder to adopt and easier to leave.

Which profile wins depends on how quickly friction falls. It is falling fast.

Pattern-by-pattern

OpenAI GPT — Λ = 0.0014, Structurally Inert

OpenAI holds the highest Standardized Rails score in the landscape (R = 0.93). Universal ecosystem support. Widest deployment vocabulary among developers. Structurally Inert anyway. Cognitive friction (Fc = 6.0) compounds under the kind of catastrophic deprecation event that occurred on February 13, 2026, when GPT-4o was retired with three months' notice. Enterprises building critical workflows on the pattern discovered the logic they had tuned was rented, not owned. **Rails without sovereignty is a trap, not an advantage.**

Anthropic Claude — Λ = 0.0059, Sub-Critical

Lower cognitive friction through ISO 42001 certification, structured safety positioning, and cleaner enterprise onboarding. The structural pattern is identical: a rented, centralized dependency. AWS Bedrock integration deepens infrastructural lock-in each quarter.

Microsoft Copilot — Λ = 0.0002, Structurally Inert

The lowest-scoring pattern in the landscape. Of 450 million commercial Microsoft 365 seats, Copilot achieves 3.3% paid conversion. The pattern layers AI onto rigid legacy user interfaces, producing friction rather than transformation. This is a sales motion sustained by Microsoft 365 distribution power, not a structural pattern.

Google Gemini — Λ = 0.0011, Structurally Inert

Embedded into BigQuery and Drive data gravity, which creates exit friction once adopted. The path to adoption is fragmented — Gemini, Vertex AI, AI Studio, and Agent Builder with no coherent signal about which deployment path to choose.

Apple Intelligence — Λ = 0.0090, Sub-Critical

The lowest Cognitive Friction score in the landscape (Fc = 2.0). Spreadability is crushed (S = 0.20) by the closed hardware requirement. The organization does not build transferable capability. It builds Apple-shaped capability.

Meta Llama — Λ = 0.0104, Sub-Critical

Previously the landscape leader under methodology version 1.0's min() β aggregation. The 2.0 geometric mean reveals structural fragility the earlier math masked. The Llama 4 EU geofencing restricted frontier access in a major enterprise market. The 700 million MAU commercial license cap remains a structural ceiling.

Mistral/DeepSeek — Λ = 0.0314, Approaching Critical

The only pattern above the 0.03 threshold. Leadership rests on balanced incentive structure: financial, regulatory, and ideological forces pulling in concert. The structural vulnerability is geopolitical. The next 18 months will determine whether geopolitical pressure or tooling maturation moves faster.

The Autonomaton — Λ = 0.0001, Structurally Inert

The pattern specification holds the landscape's highest structural base (S × R = 0.77) — 57% higher than the nearest competitor. CC BY 4.0 licensing. Model-agnostic architecture. The score does not flatter the sponsoring organization. It describes a structurally superior pattern that has not yet activated.

The operator's brief

Four directives fall out of the structural analysis.

1. Stop renting your cognition

Relying on centralized APIs for core enterprise workflows is not buying capability. It is renting logic the vendor retires on the vendor's schedule. The GPT-4o deprecation in February 2026 broke production workflows at every enterprise that had tuned prompt architectures to that specific model. Build against abstractions that outlive any single model.

2. Freeze the per-seat AI upgrade

Microsoft Copilot is the lowest-scoring pattern for a measurable reason. A 3.3% paid conversion across 450 million commercial seats is what happens when AI is layered onto rigid legacy interfaces at a $30-per-seat monthly price. If the AI seat does not produce a defensible return, it does not belong in the renewal.

3. Pivot toward sovereignty before the economics force it

At 2 million daily tokens or higher, running open-weight models on owned or rented compute produces 80 to 90 percent cost savings against the centralized APIs. METR's 89-day capability doubling time is compressing frontier performance onto commodity hardware faster than any enterprise procurement cycle can track. **The organizations that deploy specialized engineering talent now will be 18 months ahead of peers who wait for the economics to force the move.**

4. Build the middleware, not the model commitment

Model capabilities are commoditizing. The CTO's architectural responsibility is no longer to pick the winning model — it is to build the middleware that makes any given model replaceable. Governance pipelines, telemetry ownership, approval gates, and declarative skill compilation belong to the organization, not the model.

The specification gap

No CC BY 4.0 pattern specification currently exists for composable, model-agnostic, governance-first self-authoring software. Existing OSS frameworks (LangChain, LangGraph, AutoGen, Semantic Kernel, CrewAI, AutoGPT, OpenAI Swarm, Haystack, LlamaIndex) are code implementations with governance assumptions baked in. Vendor-published pattern catalogues (Google ADK, Salesforce Agentforce, Anthropic MCP, A2A) are locked to issuing vendor's platform or narrow in scope.

The Autonomaton is one attempt to fill this category. There should be others.

Conflict of interest

The Grove Foundation publishes this framework and champions the Autonomaton architecture. The Autonomaton pattern is scored using the same methodology applied to all other patterns. The Autonomaton currently scores Λ = 0.0001 — the lowest in the landscape. If Grove were tilting the methodology to favor the pattern it publishes, that score would not appear in the table. Every input variable and sub-score is cited to source. The methodology is CC BY 4.0.

About this research

The Grove Foundation publishes structural analysis of AI infrastructure adoption under Creative Commons Attribution 4.0. This is the first quarterly Λ landscape audit. 96 sources. 8 patterns. 4 historical calibrations.

The next issue examines a contractual blind spot. Every enterprise AI contract contains "telemetry" language, but the word is undefined terrain. AI vendor contracts signed in the summer of 2026 without a fulsome definition of telemetry are not subscriptions. They are permanent transfers of behavioral intelligence to the vendor that captured it under terminology that was never nailed down.

The Grove Foundation

"Design is philosophy expressed through constraint."

 · March 2026 · Issue 003 · 96 sources · 8 patterns · 4 historical calibrations